

## THOMSON TRI-TOWNSHIP, IL, USA

WMO: 722204

Lat: 42.046N

Lon: 90.108W

Elev: 616

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04996

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.4       | -0.2 | -14.6                         | 2.5 | -4.9 | -8.9 | 3.5 | 0.4  | 23.7                  | 23.4 | 20.6 | 25.9 | 4.0                   | 290  | 0.506 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.5                   | 91.1            | 74.0 | 88.9 | 73.2 | 86.1 | 71.9 | 77.6                | 87.3 | 76.0 | 85.0 | 74.4 | 82.8 | 9.2                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.9                            | 133.7 | 82.8 | 73.0 | 125.2 | 81.0 | 71.9 | 120.7 | 80.3 | 41.4          | 88.0 | 39.8 | 84.7 | 38.3 | 82.7 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |    |    |
| 20.9              | 18.6 | 16.7 | -17.3                      | 95.5 | 8.8                | 3.4 | -23.7                                              | 97.9 | -28.9      | 99.9 | -33.8      | 101.8 | -40.3      | 104.2 |    |    |
|                   |      |      | -17.6                      | 80.8 | 8.4                | 1.7 | -23.6                                              | 82.0 | -28.5      | 83.0 | -33.2      | 84.0  | -39.3      | 85.2  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 49.9   | 22.0  | 23.8  | 38.8  | 50.6  | 61.9  | 70.9  | 73.6  | 71.2  | 65.7  | 52.3  | 39.6  | 27.3  |
|                                                                                  | DBStd       | 20.57  | 12.49 | 11.51 | 11.47 | 9.24  | 8.93  | 5.86  | 5.86  | 5.44  | 7.74  | 9.33  | 9.63  | 10.88 |
|                                                                                  | HDD50       | 3218   | 867   | 734   | 380   | 104   | 10    | 0     | 0     | 0     | 2     | 83    | 334   | 704   |
|                                                                                  | HDD65       | 6362   | 1332  | 1154  | 814   | 439   | 169   | 15    | 4     | 11    | 85    | 408   | 763   | 1168  |
|                                                                                  | CDD50       | 3198   | 0     | 0     | 34    | 121   | 379   | 626   | 732   | 659   | 472   | 154   | 20    | 1     |
|                                                                                  | CDD65       | 869    | 0     | 0     | 3     | 7     | 73    | 191   | 271   | 204   | 105   | 15    | 0     | 0     |
|                                                                                  | CDH74       | 8177   | 0     | 0     | 25    | 113   | 719   | 1736  | 2683  | 1785  | 971   | 145   | 0     | 0     |
|                                                                                  | CDH80       | 2726   | 0     | 0     | 3     | 19    | 216   | 573   | 1017  | 560   | 308   | 30    | 0     | 0     |
| Wind                                                                             | WSAvg       | 7.1    | 7.6   | 7.6   | 8.3   | 9.3   | 7.7   | 6.4   | 5.2   | 5.0   | 5.8   | 7.0   | 7.8   | 7.4   |
| Precipitation                                                                    | PrecAvg     | 36.3   | 1.4   | 1.5   | 2.2   | 3.5   | 4.3   | 5.1   | 3.7   | 4.3   | 3.5   | 2.6   | 2.6   | 1.6   |
|                                                                                  | PrecMax     | 47.6   | 3.2   | 3.6   | 5.2   | 6.7   | 8.6   | 10.4  | 7.1   | 7.5   | 6.4   | 5.1   | 6.3   | 4.2   |
|                                                                                  | PrecMin     | 26.9   | 0.4   | 0.2   | 0.6   | 1.4   | 0.7   | 1.5   | 0.7   | 1.1   | 0.9   | 0.5   | 0.0   | 0.3   |
|                                                                                  | PrecStd     | 4.8    | 0.7   | 1.0   | 1.6   | 1.5   | 1.9   | 2.4   | 1.6   | 1.9   | 1.5   | 1.5   | 1.7   | 0.9   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.9  | 54.6  | 77.7  | 82.3  | 90.8  | 93.1  | 97.4  | 92.6  | 91.5  | 83.9  | 70.3  | 58.6  |
|                                                                                  |             | MCWB   | 45.8  | 47.3  | 63.5  | 60.7  | 70.8  | 74.5  | 77.6  | 73.1  | 74.2  | 68.7  | 58.4  | 54.6  |
|                                                                                  | 2%          | DB     | 45.5  | 46.2  | 67.9  | 77.3  | 85.6  | 90.0  | 91.4  | 88.5  | 87.9  | 78.5  | 63.5  | 50.0  |
|                                                                                  |             | MCWB   | 42.1  | 41.4  | 57.3  | 60.1  | 67.3  | 72.7  | 75.3  | 74.0  | 72.1  | 65.3  | 54.5  | 46.7  |
|                                                                                  | 5%          | DB     | 40.9  | 41.4  | 62.5  | 72.1  | 81.7  | 86.3  | 88.9  | 85.9  | 83.2  | 72.6  | 58.7  | 45.3  |
|                                                                                  |             | MCWB   | 37.8  | 37.1  | 53.3  | 56.9  | 65.1  | 71.4  | 74.1  | 72.5  | 69.1  | 62.0  | 52.1  | 42.5  |
|                                                                                  | 10%         | DB     | 37.1  | 37.5  | 55.3  | 67.7  | 77.9  | 82.4  | 85.8  | 82.4  | 80.4  | 67.9  | 54.2  | 41.0  |
|                                                                                  |             | MCWB   | 35.1  | 34.8  | 47.9  | 54.8  | 64.0  | 69.5  | 73.3  | 70.3  | 67.4  | 60.0  | 48.4  | 38.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 48.5  | 47.8  | 63.8  | 65.4  | 73.8  | 78.7  | 81.1  | 78.3  | 76.5  | 72.2  | 60.7  | 55.7  |
|                                                                                  |             | MCDB   | 50.0  | 51.7  | 74.4  | 74.9  | 86.7  | 89.3  | 90.0  | 86.6  | 86.7  | 81.7  | 66.5  | 57.8  |
|                                                                                  | 2%          | WB     | 41.6  | 41.9  | 59.4  | 62.0  | 71.0  | 75.6  | 78.3  | 76.5  | 74.2  | 68.3  | 56.5  | 46.9  |
|                                                                                  |             | MCDB   | 45.1  | 46.5  | 66.7  | 72.1  | 81.2  | 85.4  | 88.2  | 84.6  | 83.1  | 74.3  | 62.0  | 48.7  |
|                                                                                  | 5%          | WB     | 38.1  | 37.7  | 54.4  | 59.4  | 68.5  | 74.0  | 76.7  | 74.6  | 72.1  | 64.8  | 52.9  | 42.4  |
|                                                                                  |             | MCDB   | 40.6  | 40.6  | 60.2  | 70.0  | 77.7  | 83.0  | 86.2  | 82.4  | 80.2  | 70.0  | 57.3  | 44.7  |
|                                                                                  | 10%         | WB     | 35.3  | 35.3  | 48.5  | 56.5  | 66.1  | 72.0  | 74.9  | 72.7  | 69.8  | 60.6  | 48.7  | 38.3  |
|                                                                                  |             | MCDB   | 37.3  | 37.5  | 55.3  | 66.0  | 74.4  | 80.5  | 83.1  | 80.1  | 76.5  | 67.1  | 53.7  | 40.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.3  | 16.0  | 17.6  | 20.8  | 21.4  | 20.3  | 20.5  | 20.4  | 22.7  | 20.0  | 16.6  | 13.7  |
|                                                                                  |             | MCDBR  | 17.9  | 18.2  | 25.8  | 29.2  | 25.8  | 23.8  | 23.1  | 23.0  | 25.4  | 26.4  | 23.4  | 18.2  |
|                                                                                  |             | MCWBR  | 14.4  | 14.1  | 15.4  | 15.4  | 12.1  | 11.0  | 9.9   | 10.5  | 11.3  | 14.4  | 15.7  | 14.9  |
|                                                                                  | 5% WB       | MCDBR  | 17.2  | 17.6  | 22.6  | 24.4  | 22.1  | 20.8  | 20.2  | 18.9  | 20.6  | 20.8  | 21.2  | 17.2  |
|                                                                                  |             | MCWBR  | 14.2  | 14.2  | 15.2  | 15.0  | 11.6  | 11.0  | 10.4  | 9.9   | 10.6  | 13.9  | 15.7  | 15.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.287  | 0.302 | 0.341 | 0.387 | 0.414 | 0.427 | 0.445 | 0.422 | 0.393 | 0.343 | 0.301 | 0.286 |       |
|                                                                                  | taud        | 2.401  | 2.372 | 2.372 | 2.257 | 2.244 | 2.253 | 2.193 | 2.256 | 2.313 | 2.450 | 2.501 | 2.429 |       |
|                                                                                  | Ebn at Noon | 274    | 287   | 287   | 279   | 274   | 270   | 263   | 266   | 266   | 268   | 268   | 264   |       |
|                                                                                  | Edh at Noon | 25     | 30    | 34    | 42    | 43    | 43    | 45    | 41    | 36    | 28    | 23    | 23    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 555    | 868   | 1184  | 1473  | 1728  | 1942  | 1931  | 1703  | 1424  | 932   | 613   | 449   |       |
|                                                                                  | RadStd      | 66     | 98    | 110   | 123   | 120   | 157   | 120   | 111   | 117   | 112   | 50    | 43    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)  | (b)                     | (c)   | (d)     | (e)    | (f)   | (g)     | (h)   | (i) | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (54 neighbors) |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page